

PRODUCT NAME	Shredder	EE TEAM
APPROVED BY		Fearghus Tyler
DATE OF APPROVAL	2026-06-01	Bao Nguyen
PART COUNT		62 Yaqoub Rabiah
TOTAL COST		\$969.88 Fox Kang

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	UNITS	LINK	UNIT COST	AMOUNT
5KE182BC205B	AC Motor (3 HP)	GE 3-phase TEFC induction motor; 230 V (low-voltage), 1750 RPM, 7.6 A FLA, NEMA 182T frame. Bench-test motor.	1	EA	https://www.electricmotorsforless.com/	\$0.00	\$0.00
HY02D211B-T	VFD	Huanyang variable frequency drive, 110 V single-phase input, 220 V 3-phase output, 2.2 kW (3 HP). Selected via decision matrix.	1	EA	https://www.amazon.com/Huanyang-1-Phase-Variable-Frequency-Output-3hp/dp/B0DMF3GB38	\$200.00	\$200.00
APCS-300R30-AD	Brake Resistor	LS Electric 300 W, 30 ohm aluminum-clad braking resistor for VFD dynamic braking on P+/PR. Purchased from AutomationDirect. Note: an earlier identical-part-number unit (ATO branded) was ordered but never arrived after shipping; replaced with this LS Electric unit which is the brake resistor actually installed.	1	EA	https://www.automationdirect.com/adc/shopping/catalog/motion_control/ls_electric_l7c_l7ca_series/servo_systems/servo_accessories/apcs-300r30-ad	\$29.00	\$29.00
71008	Main Power Switch	Powertec 120 V ON/OFF main disconnect switch for entire control panel.	1	EA	https://powertecproducts.com/71008/	\$15.00	\$15.00
TMC 8 3C 15A (2907618)	VFD Branch Breaker	Phoenix Contact thermal-magnetic DIN-rail circuit breaker, 15 A, on VFD neutral leg (per as-built). Obtained at no cost from the PSU EPL free pile.	1	EA	https://www.phoenixcontact.com/	\$0.00	\$0.00
NDR-480-24	24 VDC Power Supply	Mean Well 480 W, 24 VDC DIN-rail power supply for PLC, HMI, indicator LEDs, and VFD control loops. Donated by sponsor / obtained at no cost to the EE team.	1	EA	https://www.meanwell.com/	\$0.00	\$0.00
SD-N35	Main Power Contactor	Mitsubishi 3-pole magnetic contactor. Energized through E-stop chain via E22B1 NC contact.	1	EA	https://www.ebay.com/itm/... (tpmachinery seller, order 23-14145-39817, Jan 2026)	\$12.59	\$12.59
EURO S4LH	Fuse Block (button common)	Morsettitalia fuse-holder terminal block. Holds 1 A fast-blow fuse protecting the shared pushbutton +24 V supply.	1	EA	https://www.morsettitalia.com/	\$8.00	\$8.00
EURO S10H-5H	Fuse Block (AC branches)	Morsettitalia fuse-holder terminal block for AC branch protection.	1	EA	https://www.morsettitalia.com/	\$12.00	\$12.00
H2-DM1E	PLC CPU / Ethernet	AutomationDirect Do-more CPU with built-in Ethernet, fits D2-09B-1 DL205 rack. Purchased Jan 2026 from planet-surplus as a USED combo for \$350 total, includes: D2-09B-1 chassis, D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2, and D2-FILL filler panels.	1	EA	https://www.ebay.com/itm/... (planet-surplus seller, order 20-14113-87209, Jan 2026)	\$350.00	\$350.00
D2-09B-1	PLC Base / Rack	AutomationDirect DirectLOGIC 205 9-slot AC-powered base. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
D2-08ND3	Discrete Input Module	AutomationDirect 8-ch, 24 VDC, sourcing/sinking discrete input module. Slot 2. Inputs X0-X7. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
F2-08AD-1	Analog Input Module	AutomationDirect 8-ch, 4-20 mA analog current input module. Slot 3. CH1+ reads VFD VO current monitor. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
F2-04RTD	RTD Input Module	AutomationDirect 4-ch RTD input module. Slot 4. Installed but unused (spare). Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
D2-08TR	Relay Output Module	AutomationDirect 8-ch relay output module. Slot 5. Drives button LEDs (Y0-Y4) and VFD FOR/REV/RST (Y6/Y7/Y5). Included in the H2-DM1E used combo purchase (eBay listing showed D2-08TD2 but seller substituted D2-08TR which is what we use).	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
F2-08DA-2	Analog Output Module	AutomationDirect 8-ch, 0-10 V analog voltage output module. Slot 6. +V1 drives VFD VI speed reference. Included in the H2-DM1E used combo purchase (eBay listing showed F2-02DA but actual unit is F2-08DA-2).	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00

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D2-FILL	Rack Filler Panels	AutomationDirect blank fill panels for unused rack slots 7-9. Included in the H2-DM1E used combo purchase.	3	EA	https://www.automationdirect.com/	\$0.00	\$0.00
EA1-T4CL	HMI Touch Panel	AutomationDirect C-More Micro-Graphic 4-inch color touch panel HMI. Purchased used from eBay for \$148.95 (Apr 2026). Purchase included the serial cable for HMI-to-CPU communication (RS-232 to H2-DM1E serial port).	1	EA	https://www.ebay.com/itm/306662618920	\$148.95	\$148.95
S2PR-P3G	Forward Pushbutton Head	Autonics 22 mm illuminated green pushbutton head (Forward command -> X0).	1	EA	https://www.autonics.com/us/model/S2PR-P3GAD	\$1.90	\$1.90
S2PR-P3Y	Reverse Pushbutton Head	Autonics 22 mm illuminated yellow pushbutton head (Reverse command -> X1).	1	EA	https://www.autonics.com/us/model/S2PR-P3YAD	\$1.90	\$1.90
S2PR-P3B	Run/Deadman Pushbutton Head	Autonics 22 mm illuminated blue pushbutton head (Run 1 -> X2, Run 2 -> X3).	2	EA	https://www.autonics.com/us/model/S2PR-P3BAD	\$1.90	\$3.80
S2PR-P3R	Reset Pushbutton Head	Autonics 22 mm illuminated red pushbutton head (planned Reset -> X6).	1	EA	https://www.autonics.com/us/model/S2PR-P3RAD	\$4.50	\$4.50
SA-CA	NO Contact Block (1NO)	Autonics finger-safe 1NO contact block. 110 VAC/10 A. Used with each S2PR pushbutton.	5	EA	https://www.autonics.com/us/model/SA-CA	\$2.80	\$14.00
SA-LDG	LED Block - Green (24 VDC)	Autonics modular LED indicator block, green, 12-24 VDC. Forward button illumination -> Y0.	1	EA	https://www.autonics.com/us/model/SA-LDG	\$4.50	\$4.50
SA-LDY	LED Block - Yellow (24 VDC)	Autonics modular LED indicator block, yellow, 12-24 VDC. Reverse button illumination -> Y1.	1	EA	https://www.autonics.com/us/model/SA-LDY	\$4.50	\$4.50
SA-LDB	LED Block - Blue (24 VDC)	Autonics modular LED indicator block, blue, 12-24 VDC. Run 1 and Run 2 illumination -> Y2, Y3.	2	EA	https://www.autonics.com/us/model/SA-LDB	\$4.50	\$9.00
SA-LDR	LED Block - Red (24 VDC)	Autonics modular LED indicator block, red, 12-24 VDC. Reset button illumination (planned).	1	EA	https://www.autonics.com/us/model/SA-LDR	\$4.50	\$4.50
E22B1	Emergency Stop Pushbutton	Cutler-Hammer / Eaton 22 mm mushroom E-stop, NC contact. In series with SD-N35 contactor coil.	1	EA	https://www.eaton.com/	\$25.00	\$25.00
Wamco 525	Fault Indicator Light	Wamco panel-mount LED indicator, 28 VDC, 0.6 W. Driven by PLC Y4 as machine-fault indicator.	1	EA	https://www.wamco.com/	\$15.00	\$15.00
Banana Socket (Blue)	Banana Socket Blue	Generic 4 mm panel-mount banana socket, blue. Quick-disconnect for VFD U output to motor T1. Obtained at no cost from a free pile.	1	EA		\$0.00	\$0.00
Banana Socket (White)	Banana Socket White	Generic 4 mm panel-mount banana socket, white. VFD V output to motor T2. Obtained at no cost from a free pile.	1	EA		\$0.00	\$0.00
Banana Socket (Red)	Banana Socket Red	Generic 4 mm panel-mount banana socket, red. VFD W output to motor T3. Obtained at no cost from a free pile.	1	EA		\$0.00	\$0.00
G1X1LG6	Slotted Wire Duct, 1"x1"	Panduit Type G wide-finger slotted PVC duct, light gray, 6 ft lengths.	6	FT	https://www.panduit.com/	\$3.43	\$20.58
C1LG6	Duct Cover, 1"W	Panduit duct cover for G1X1LG6, light gray, 6 ft lengths.	6	FT	https://www.panduit.com/	\$1.20	\$7.20
18 AWG Blue	Control Wire, Blue	18 AWG stranded blue control wire (NFPA 79 §13.2 DC control). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
18 AWG White	Control Return Wire	18 AWG stranded white wire for DC return (NFPA 79 §13.2). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
12 AWG Black	Power Wire, Black	12 AWG black wire for AC mains and motor leads. Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
12 AWG Green/Yellow	Ground Wire	12 AWG green/yellow stranded equipment grounding wire (NFPA 79). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
GMA-500mA or 0217.500MXP	Fuse - 500 mA F	Bussmann or Littelfuse 5x20 mm fast-blow fuse, 500 mA, 250 V. (Currently not installed per as-built note - retained as recommended addition.)	1	EA		\$1.50	\$1.50
GMA-1A or equiv.	Fuse - 1 A F	5x20 mm fast-blow fuse, 1 A, 250 V. Protects shared +24 V pushbutton common (in EURO S4LH).	1	EA		\$1.50	\$1.50
LM2596	DC-DC Buck Converter (24V to 5V/3.3V)	Seloky LM2596-based adjustable DC-DC buck converter module with voltmeter display (sold as 2-pack). Steps down 24 VDC panel supply to the input voltage required by the capacitive touch module.	1	Pack	https://www.amazon.com/Converter-Adjustable-Regulator-Voltmeter-Display/dp/B0F1M2SR31	\$7.99	\$7.99

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SEB001	Control Panel Enclosure	Vorksmen industrial control enclosure (steel) housing the entire electrical system including PLC rack, VFD, contactor, PSU, terminal blocks, and operator panel.	1	EA	https://www.amazon.com/	\$59.98	\$59.98
AE14921a	3V Relay	AEDIKO 3 VDC PCB-mount relay. Used in the capacitive touch module sub-circuit to interface the low-voltage touch sensor output with the 24 VDC panel control voltage.	1	EA	https://www.amazon.com/	\$6.99	\$6.99
DGN-001	<i>Bao's Dignity</i>	<i>Lost on the procurement loop after lowballing roughly every vendor between Portland and Beaverton. Quoted "open-box" / "educational discount" / "the next one is on us" until even PowerTec stopped replying. Non-returnable. Replacement TBD; budget is hopefully zero.</i>	1	EA		\$0.00	\$0.00
		TOTAL PARTS	62		TOTAL		\$969.88

PART NUMBER	PART NAME	DESCRIPTION	FROM BOM FILE	REASON NOT USED
PE145TC-2-4	Motor (alternate)	Original BOM candidate motor	BOM_02_20_26	Not used; GE 5KE182BC205B used instead for bench testing
NBE7	Circuit Breaker	Original BOM candidate breaker	BOM_1_29 / BOM_02_20_26	Never purchased; replaced by Phoenix Contact TMC 8 3C 15A
(30 A fuse, TBD)	30 A Branch Fuse	Per old BOM, 2x 30 A fuses budgeted	BOM_1_29	Never needed in final design
1866-1441-ND	DC Power Supply	120/24 VDC PSU from Digi-Key	BOM_1_29	Superseded by Mean Well NDR-480-24
(generic)	VEVOR 3HP 120V VFD	Hobby-grade VFD option	VFD_120V_3HP_Decision_Matrix	Rejected during selection (hobby-grade, limited documentation)
(generic)	ATO 3HP 120V VFD	Higher-quality VFD option	VFD_120V_3HP_Decision_Matrix	Too expensive (\$450-500) for project budget
(generic)	110V 20A 3HP VFD	Mid-tier VFD option	VFD_120V_3HP_Decision_Matrix	Huanyang HY02D211B-T selected instead
(WEG single-phase 3HP)	WEG Single Phase Motor	1-phase general purpose 3 HP motor	Motor Spec	Considered but GE 5KE182BC205B used
NT3-18-213T	Worldwide Electric Motor	3-phase TEFC 3 HP motor	Motor Spec	Considered but not purchased
(VEVOR 5HP 230V)	VEVOR High Torque Motor	5 HP 1-phase compressor-style motor	Motor Spec	Considered but not purchased
L510 (2HP / 3HP)	TECO L510 VFD	VFD comparison option	Motor Spec	Considered but Huanyang selected
G0532	Marathon Gearmotor	Integrated gearbox motor	Motor Spec	Considered but not purchased
KBWC-15K	KB Electronics speed control	Triac speed control fan-style	Motor Spec	VFD approach used instead
TTP223B	Capacitive Touch Sensor	Single-channel cap touch	BOM_1_30_beta (safety draft)	Does not meet industrial standards
CAP1293-1-SN	Capacitive Touch Chip	3-channel cap touch chip	BOM_1_30_beta (safety draft)	Chip only; needs interface board
1864-1032-ND	Capacitive Proximity Sensor	Cap proximity sensor	BOM_1_30_beta (safety draft)	Too expensive (\$109.25)
F3S-TGR-NMPC-21-10	Omron Magnetic Safety Sensor	Non-contact magnetic safety switch	BOM_1_30_beta (safety draft)	Too expensive; out of stock
RSS 36-I2-D-R-ST5	Schmersal RFID Safety Sensor	RFID safety sensor	BOM_1_30_beta (safety draft)	Suspected industrial-only / non-retail
F3SG-SR / PG	Omron Safety Light Curtain	Safety light curtain	BOM_1_30_beta (safety draft)	Out of scope for capstone budget
(Ichor-provided)	Red/Green Indicator Lights	Sponsor-provided lights	BOM_02_20_26	Not used; Wamco 525 fault light and button LEDs cover indication
(Ichor-provided)	E-stop Pushbutton	Sponsor-provided E-stop	BOM_02_20_26	Replaced with Cutler-Hammer/Eaton E22B1 procured by team
V30AE000055	Insulated Ferrules, 12 AWG (gray)	Z+F crimp ferrules, 12 AWG, 10 mm length, gray (DIN color code). 500-piece pack.	URS Electronics invoice (purchased but not used)	Purchased from URS Electronics but ultimately not used in the as-built panel. Spare stock retained for future projects.